

Modal Triplet Theory and Nonrelativistic Quantum Mechanics:

A Coherent-Sector Reconstruction and the Born-Source Boundary

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Abstract

This paper asks what Modal Triplet Theory (MTT) presently establishes about nonrelativistic quantum mechanics. We separate four questions that are often conflated: whether MTT can encode a quantum model, whether a coherent sector reconstructs its Hilbert-space dynamics, whether one selected MTT source emits the required operators and instruments, and whether the probability law is derived for every apparatus. A complete quantum-mechanical record is defined, including the complex Hilbert space, state cone, operator algebra, self-adjoint Hamiltonian and domain, unitary evolution, clock convention, composition rule, effects, instruments, and source certificates. The main theorem is conditional: if one selected upper MTT state emits this record and the evaluation diagram commutes, then MTT reconstructs the corresponding quantum theory on the declared domain. Projection alone does not create a complex structure, noncommutative observable algebra, self-adjoint Hamiltonian, tensor product, or probability law. We give sufficient closed-form and reducing-subspace conditions for an exact coherent-sector Hamiltonian, and we show why reproducing an arbitrarily supplied Schrodinger potential is an expressiveness result rather than a prediction. Time is treated as an evolution or clock parameter; the valid Mandelstam–Tamm relation replaces the former unsupported universal self-adjoint time operator. Gleason- and Busch-type theorems characterize probability assignments after Hilbert/effect structure and additivity are assumed; they do not source those assignments. A stronger positive result is available on one selected domain: the canonical $q = 79$ finite-symbol Cauchy quantum model fixes its reduced semigroup and minimal dilation, and its binary one-anchor recorder gives an exact stopped output measure and second-moment Born descent on its commuting Fock output algebra, with no fitted probability or additional classical noise. General apparatus contexts, finite-bandwidth and non-Markov control, pre-quantum probability semantics, and objective single-history selection remain open. The result is therefore a rigorous coherent-sector reconstruction with one exact selected operational model, not a complete first-principles derivation of all quantum mechanics. Contextual imports explain the exact projector and strain compilers, conditional physical-Hessian interface, free local field net, and formal quantum master equation without promoting them to a selected interacting continuum theory.

Revision note: Version 5

- Supersedes** Version 4 as the current manuscript; its release identity and historical revision note are retained below.
- Reason** The selected canonical quantum model and the later geometric and field-theoretic refinements require contextual explanation. Version 4 also misstated what bounded pullback preserves.
- Resolution** This revision integrates thirteen frozen source records, distinguishes compiler theorems from physical source selection, and corrects the pullback, no-count-amplitude, and

quasifree-convexity statements at their precise scopes. Source prose and finite checks are evidence, not independent verification of every theorem recorded in a packet.

Retained The conditional reconstruction, standard theorem ownership, closed canonical operational model, exact binary output law, finite/local field results, and existential hidden HYM result remain intact.

Open boundary

General apparatuses and actualization, the selected physical HYM-to-operator map, and a positive fixed-coupling interacting continuum theory are not supplied. A pure nonnegative repair cost is not identified with the physical signed action.

Revision note: Version 4

Supersedes Version 4 supersedes Version 3 and its claims of a complete first-principles derivation of quantum mechanics, a universal MTT derivation of the Born rule, and automatic realization of arbitrary Schrodinger dynamics from the former ten-dimensional tri-product geometry.

Reason The former paper combined valid standard operator theory with unsupported source claims. It did not distinguish an assumed complex Hilbert space from one derived by projection, treated bounded pullback as sufficient for closedness of an unbounded quadratic form, and reproduced an arbitrary target potential by inserting that same potential upstairs. It also used a self-adjoint time operator without establishing one, treated Gleason-type probability characterization as a physical source theorem, and inferred general POVMs, entanglement, and Lindblad dynamics from projection alone. The literal three-factor internal product used there is no longer the canonical MTT geometry.

Resolution This revision defines the complete lower quantum record and a typed, conditional reconstruction theorem. It states explicit hypotheses for complex, symplectic, self-adjoint, unitary, clock, composition, and instrument structures; separates exact compression from approximate elimination; replaces the time-operator claim by operational clock and Mandelstam–Tamm statements; and separates Gleason–Busch characterization from probability sourcing. Standard mathematical theorems retain their literature ownership.

Retained content

Closed semibounded forms remain an appropriate route to self-adjoint Hamiltonians once their domains and closedness are proved. Stone’s theorem then supplies unitary evolution. Target-form matching remains a useful exact representation test. Quantum instruments, dilations, tensor products, semiclassical kernels, and weak-coupling master equations remain valid lower-level tools under their standard hypotheses. The canonical $q = 79$ binary recorder adds a selected exact output-measure result on its declared domain.

Remaining boundary

The universal Born-source blocker remains open. The current exact result covers one selected binary P/Q counting context with standard normal-state operational

semantics. It does not yet cover arbitrary preparations and apparatuses, finite-bandwidth or non-Markov detectors, pre-quantum probability semantics, or selection of one uniquely actual history.

Contents

Revision note: Version 4	2
1 The question in its correct order	4
2 The complete quantum record	5
3 Conditional coherent-sector reconstruction	6
4 Complex, symplectic, and noncommutative structure	7
5 Hamiltonians from forms and exact compression	7
5.1 The valid quadratic-form route	7
5.2 When a compression is exact	8
6 Representation is not prediction	9
7 Unitary dynamics, clocks, and uncertainty	9
7.1 Autonomous and time-dependent dynamics	9
7.2 Time is not automatically an observable	9
7.3 Ordinary observable uncertainty	10
8 Three probability questions	10
8.1 What Gleason and Busch establish	10
8.2 Effects and instruments	11
9 The selected $q = 79$ binary recorder	11
9.1 Input data	11
9.2 Exact output law	12
9.3 What this closes and what it does not	13
9.4 Consumer boundary: continuum transport and ontology	13
10 Composition, entanglement, and locality	13
11 Open systems and alternative representations	14
11.1 Weak-coupling master equations	14
11.2 Path integrals and semiclassical kernels	14
12 Two concrete tests	14
12.1 A two-level recorder	14
12.2 A harmonic oscillator embedding	15

13 From a geometric source to a retained operator	15
13.1 Projector realization and its domains	15
13.2 Strain coordinates and the cost of eliminating modes	16
13.3 What changes when an edge vanishes	16
13.4 Monodromy, the shared root, and operator symmetry	17
13.5 Which source packages are still needed	17
14 The local field interface and its limits	18
14.1 A selected free net is already available	18
14.2 Local quantum master equations are formal interacting results	18
14.3 What the later regulator refinements actually control	19
14.4 Free positivity and fixed-coupling nonpromotion	19
15 Relation to the current MTT program	19
16 Claim disposition	20
17 Completion contract	21
18 Discussion	22
18.1 What has actually been achieved	22
18.2 Why the narrower statement is stronger science	22
18.3 Best next theorem	22
19 Conclusion	22

1 The question in its correct order

Nonrelativistic quantum mechanics is not one equation. It is a coordinated mathematical structure. A state belongs to a complex Hilbert space or a density-operator cone; observables are represented by a generally noncommutative operator algebra; dynamics is generated by a self-adjoint Hamiltonian; composition uses a tensor product; and experiments are described by effects or instruments. These pieces constrain one another, but no one of them is identical to the others [1, 2, 3].

MTT begins with an upper state, admissibility conditions, reduction maps, and coherent or fixed sectors [4]. It is therefore natural to ask whether a coherent sector can carry quantum mechanics. Four logically distinct questions must be kept apart:

1. *Encoding.* Can a chosen quantum model be represented inside an MTT carrier?
2. *Reconstruction.* If the required lower data are present, does MTT evaluation reproduce the same states, dynamics, and experiments?
3. *Selection.* Does one selected MTT source emit those lower data without inserting the target theory by hand?
4. *Universality.* Does the same source construction cover every allowed preparation, apparatus context, and limiting regime?

An exact answer to the first question can coexist with an open answer to the third. This is what happens when an arbitrary potential V is placed upstairs and then recovered downstairs: the

construction proves expressive capacity, not why nature selected V . Conversely, a selected result on one detector context is genuine source progress even when the universal apparatus theorem remains open.

The paper's positive claims occupy two levels. The general theory is a conditional reconstruction. The selected $q = 79$ finite-symbol Cauchy model also supplies a complete operational model on its declared domain, including its reduced dynamics, minimal dilation, and binary probability and capture result. The conditional formulation is not a reopening of this established model. Sections 13 and 14 explain which further source and field constructions extend that domain and which remain conditional.

2 The complete quantum record

It is useful to say explicitly what must be supplied before a lower quantum theory exists. This prevents a projector, a spectrum, or a formal Schrodinger equation from silently standing in for the entire theory.

Definition 2.1 (Nonrelativistic quantum record). A nonrelativistic quantum record is a tuple

$$\mathcal{R}_{\text{QM}} = (\mathcal{H}, \mathcal{S}, \mathcal{A}, \mathcal{D}, H, \hbar, U, \mathcal{T}, \mathcal{E}, \mathcal{I}, \boxtimes, \mathcal{L}, \mathcal{X})$$

with the following typed rows.

1. $\mathcal{H}$ is a complex separable Hilbert space.
2. $\mathcal{S}$ is a declared state space, for example the positive trace-class operators ρ with $\text{Tr } \rho = 1$.
3. $\mathcal{A}$ is a unital C^* - or von Neumann observable algebra represented on $\mathcal{H}$; unbounded observables are supplied with domains and affiliation data.
4. $\mathcal{D} \subset \mathcal{H}$ is the dense form or operator domain needed below, and H is a self-adjoint Hamiltonian, normally bounded below.
5. $\hbar > 0$ is the action normalization and $U(t) = \exp(-itH/\hbar)$ is the strongly continuous unitary evolution.
6. $\mathcal{T}$ declares how time is used: as an external evolution parameter, as readings of a physical clock, or through a specified covariant clock POVM.
7. $\mathcal{E}$ contains the effects or POVMs used by the allowed experiments, while $\mathcal{I}$ contains their completely positive instruments and record spaces.
8. $\boxtimes$ is the composition rule, ordinarily $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$, together with the corresponding local algebras.
9. $\mathcal{L}$ records locality, symmetry, superselection, and boundary conditions appropriate to the model.
10. $\mathcal{X}$ contains source and error certificates: which rows are selected by MTT, which are imported lower data, the domain on which the comparison is exact, and any residual or approximation bound.

The record is deliberately larger than $(\mathcal{H}, H)$. A Hamiltonian does not specify which instruments exist. A POVM does not specify the post-record state change. A real inner-product space does not yet specify multiplication by i . A tensor-product notation does not prove that the MTT source map respects subsystem locality.

Let $\mathcal{U}_{\text{adm}}$ be the declared upper MTT domain. A partial source map

$$\mathcal{S}_{\text{QM}}: \mathcal{U}_{\text{adm}} \dashrightarrow \{\text{quantum records}\}$$

is allowed to be undefined when one or more rows cannot be emitted. This is preferable to filling a missing row with the desired lower object and calling the result a derivation.

3 Conditional coherent-sector reconstruction

Suppose an upper MTT state z has a coherent sector $\mathcal{H}_{\text{coh}}(z)$, and let

$$\mathcal{P}_z: \mathcal{H}_{\text{up}}(z) \longrightarrow \mathcal{H}_{\text{coh}}(z)$$

be its orthogonal coherent projector. The notation alone says only that $\mathcal{P}_z^2 = \mathcal{P}_z = \mathcal{P}_z^*$. It does not say that the coherent range is complex, invariant under an upper Hamiltonian, closed under a chosen observable algebra, or equipped with a probability interpretation.

Let $\text{Eval}_{\text{MTT}}(z, C)$ denote the MTT prediction for an allowed preparation-and-apparatus context C . Let $\text{Eval}_{\text{QM}}(\mathcal{R}, C)$ denote the standard quantum evaluation of the record in Definition 2.1.

Theorem 3.1 (Conditional MTT-QM reconstruction). *Fix an upper state $z \in \mathcal{U}_{\text{adm}}$ and a declared family $\mathfrak{C}_z$ of contexts. Assume:*

- (R1) $\mathcal{S}_{\text{QM}}(z) = \mathcal{R}_{\text{QM}}(z)$ is defined and every row of Definition 2.1 needed by $\mathfrak{C}_z$ is present;
- (R2) the Hilbert identification $J_z: \mathcal{H}_{\text{coh}}(z) \rightarrow \mathcal{H}$ is unitary and intertwines the declared state and observable maps;
- (R3) coherent evolution is exact, or carries a stated error:
$$J_z \mathcal{P}_z U_{\text{up}}(t) \iota_z = U(t) J_z + \mathcal{E}_t, \quad \|\mathcal{E}_t\| \leq \varepsilon_t;$$
- (R4) every allowed upper instrument descends to the corresponding lower instrument, again exactly or with a declared norm bound; and
- (R5) MTT evaluation factors through the lower record:

$$\text{Eval}_{\text{MTT}}(z, C) = \text{Eval}_{\text{QM}}(\mathcal{R}_{\text{QM}}(z), C) + r_C$$

with $|r_C| \leq \delta_C$.

Then MTT reconstructs the quantum model $\mathcal{R}_{\text{QM}}(z)$ on $\mathfrak{C}_z$, exactly when $\varepsilon_t = \delta_C = 0$, and with the displayed control otherwise.

Proof. The source map supplies a well-typed lower record by (R1). Conditions (R2) and (R3) transport states, observables, and dynamics to the lower Hilbert space. Condition (R4) transports the operational contexts, including their record probabilities and conditional states. Substitution into (R5) gives the asserted equality or error bound for every context in $\mathfrak{C}_z$. No claim outside that context family follows. $\square$

The theorem is intentionally simple. Its value is diagnostic. It turns the phrase “MTT gives quantum mechanics” into a finite list of diagrams that can succeed or fail separately. It also distinguishes a theorem of reconstruction from a theorem that selects the record’s numerical or operator entries.

4 Complex, symplectic, and noncommutative structure

A complex Hilbert space can be viewed as a real Hilbert space equipped with a compatible complex structure. If g is a real inner product and $J^2 = -\text{Id}$ with J orthogonal, then

$$\omega(x, y) = g(Jx, y)$$

is symplectic, and g , J , and ω determine the corresponding complex inner product after a sign convention is fixed. Conversely, on a complex Hilbert space, multiplication by i supplies J , while the real and imaginary parts of the inner product supply g and ω .

This equivalence explains why Hilbert and symplectic descriptions can agree. It does not explain where J comes from. A real orthogonal projector P has eigenvalues 0 and 1; by itself it does not define an operator squaring to $-\text{Id}$. A selected MTT source must therefore emit a complex line, root-plane complex structure, polarization, or equivalent datum and prove that it descends to the same coherent sector.

The same caution applies to noncommutativity. Compression of an already noncommutative algebra can preserve noncommutativity, but projection alone does not generate the canonical commutation relations. If $\mathcal{A}_{\text{up}} \subset \mathcal{B}(\mathcal{H}_{\text{up}})$, the compressed operators

$$A_{\text{coh}} = \mathcal{P}AP|_{\text{Ran } \mathcal{P}}$$

need not even form an algebra, because

$$(\mathcal{P}AP)(\mathcal{P}BP) \neq \mathcal{P}ABP$$

unless suitable invariance conditions hold. The observable-algebra row in Definition 2.1 is therefore an independent obligation, not a consequence of the word “coherent.”

5 Hamiltonians from forms and exact compression

5.1 The valid quadratic-form route

Closed semibounded quadratic forms are a robust way to construct self-adjoint Hamiltonians. The first representation theorem associates a unique self-adjoint lower-bounded operator with every densely defined closed semibounded form; KLMN controls suitable form-bounded perturbations [2, 1]. These are imported standard theorems. MTT must still prove their hypotheses for the selected form.

There are three different issues: closedness of a pullback form, density of its domain, and identification of its operator with a reducing operator part. For a bounded map I and a closed semibounded form q , the pullback $q(Ix, Iy)$ on the *full preimage* $I^{-1}\mathcal{D}(q)$ is closed and semibounded. Indeed, after shifting q to a nonnegative form, a sequence Cauchy in $\|x\|^2 + q_\beta(Ix)$ converges downstairs and in the upstairs form norm; boundedness of I identifies the two limits. This norm is equivalent to a sufficiently shifted pullback form norm. Density need not follow: on ℓ^2 , take $q(v) = \sum_{n \geq 1} n^2 |v_n|^2$ and the bounded map $I : \mathbb{C} \rightarrow \ell^2$, $Ic = (c/n)_{n \geq 1}$. Its preimage form domain is $\{0\}$. Thus the historical Version 4 objection to *closedness* was incorrect; a densely defined Hamiltonian and an exact operator restriction still require the additional hypotheses below.

Proposition 5.1 (Closed reducing pullback). *Let q_{up} be a densely defined closed semibounded form on $\mathcal{H}_{\text{up}}$. Let $I : \mathcal{H}_{\text{coh}} \rightarrow \mathcal{H}_{\text{up}}$ be an isometry with closed range M , let P_M be the orthogonal projector onto M , and assume*

$$P_M \mathcal{D}(q_{\text{up}}) \subset \mathcal{D}(q_{\text{up}}).$$

Assume also that

$$I\mathcal{H}_{\text{coh}} \cap \mathcal{D}(q_{\text{up}}) = I\mathcal{D}_{\text{coh}},$$

and assume M reduces the form:

$$q_{\text{up}}(x, y) = 0 \quad \text{for } x \in M \cap \mathcal{D}(q_{\text{up}}), \quad y \in M^\perp \cap \mathcal{D}(q_{\text{up}}).$$

Then

$$q_{\text{coh}}(\psi, \phi) = q_{\text{up}}(I\psi, I\phi), \quad \psi, \phi \in \mathcal{D}_{\text{coh}},$$

is densely defined, closed, and semibounded whenever $\mathcal{D}_{\text{coh}}$ is dense. Its represented Hamiltonian is unitarily equivalent to the part of the upper Hamiltonian on M .

Proof. The isometry preserves the Hilbert norm and transports the form norm on $\mathcal{D}_{\text{coh}}$ to the form norm on $M \cap \mathcal{D}(q_{\text{up}})$. A form-Cauchy sequence downstairs therefore maps to a form-Cauchy sequence upstairs. Closedness of q_{up} , closedness of M , and the domain identity return a limit in $I\mathcal{D}_{\text{coh}}$. This proves closedness downstairs. Semiboundedness is inherited through the isometry. Form reduction gives an orthogonal form sum on $M \oplus M^\perp$, so the representation theorem identifies the lower operator with the upper operator part on M . $\square$

The proposition is a sufficient theorem, not a claim that every coherent projector satisfies its assumptions. A general cost pullback need not have a dense domain or identify an operator part. A nonreducing range couples retained and discarded modes even when its pullback form is closed.

5.2 When a compression is exact

Let H_{up} be self-adjoint and P an orthogonal projector. If P commutes with every spectral projection of H_{up} , then $M = \text{Ran } P$ reduces H_{up} . The restriction

$$H_M = H_{\text{up}}|_{\mathcal{D}(H_{\text{up}}) \cap M}$$

is self-adjoint on M , and

$$P e^{-itH_{\text{up}}/\hbar}|_M = e^{-itH_M/\hbar}.$$

This is exact coherent-sector dynamics.

Without reduction, PHP is not automatically self-adjoint on the naively compressed domain, and $P e^{-itH_{\text{up}}/\hbar} P$ need not be a unitary group on M . In block form,

$$H_{\text{up}} = \begin{pmatrix} H_{MM} & V \\ V^* & H_{\perp\perp} \end{pmatrix},$$

the coupling V feeds discarded modes back into the coherent sector. At a resolvent parameter z , an exact elimination, where defined, uses the Schur–Feshbach expression

$$H_{\text{eff}}(z) = H_{MM} - V(H_{\perp\perp} - z)^{-1}V^*.$$

It is energy dependent and is not the same object as H_{MM} . A genuine approximation must bound the resolvent, the coupling, and the relevant time or spectral window. This is why “project and evolve” is not a universal derivation of unitary lower dynamics.

6 Representation is not prediction

The old reconstruction theorem allowed an arbitrary target

$$H_{\text{target}} = -\frac{\hbar^2}{2m}\Delta + V$$

and chose MTT background or boundary data containing the same V . Under appropriate Kato or form assumptions, this can be made mathematically exact. Its logical status is nevertheless representational.

Proposition 6.1 (Exact target embedding). *Let q_{target} be any densely defined closed semibounded form on $\mathcal{H}$. Let $q_{\perp}$ be such a form on an auxiliary Hilbert space $\mathcal{K}$. On $\mathcal{H}_{\text{up}} = \mathcal{H} \oplus \mathcal{K}$, define*

$$q_{\text{up}} = q_{\text{target}} \oplus q_{\perp}$$

and let P project onto $\mathcal{H}$. Then the coherent restriction of q_{up} is exactly q_{target} , and its represented Hamiltonian is H_{target} .

Proof. The direct sum is closed and semibounded on the direct-sum form domain. The first summand reduces it. Applying Proposition 5.1 to the canonical inclusion gives the result. $\square$

This proposition is useful. It proves that the MTT carrier is not too small to represent the target class. It can also test a proposed decoder and normalization. It does not select m , V , the boundary condition, or $\hbar$, because all of them entered the source record. Prediction begins only when an upstream theorem emits those entries from data that did not already contain the desired answer.

7 Unitary dynamics, clocks, and uncertainty

7.1 Autonomous and time-dependent dynamics

For a self-adjoint H , Stone's theorem gives the strongly continuous unitary group

$$U(t) = e^{-itH/\hbar}.$$

Thus unitarity follows after self-adjointness and the time parameter have been established, not from projection alone [5, 1]. For a time-dependent family $H(t)$, a propagator requires common-domain, stability, and regularity hypotheses of the appropriate Kato theorem [2]. Writing $H(t)$ is not by itself an existence proof.

7.2 Time is not automatically an observable

In standard nonrelativistic mechanics, t usually labels evolution. It is not therefore represented by a universal self-adjoint operator conjugate to every semibounded Hamiltonian. If a physical clock is part of an experiment, the clock row $\mathcal{T}$ may instead contain a covariant POVM E_T satisfying a declared covariance convention such as

$$U(s)E_T(B)U(s)^* = E_T(B + s).$$

The clock system, its coupling, resolution, and calibration must then be part of the source record.

The valid energy–time statement used here is operational. For a time-independent observable A , a state in the necessary domains, and

$$\tau_A = \frac{\Delta_\psi A}{\left| \frac{d}{dt} \langle A \rangle_\psi \right|},$$

the Robertson inequality applied to A and H , together with the Heisenberg equation, gives

$$\tau_A \Delta_\psi H \geq \frac{\hbar}{2}.$$

This is the Mandelstam–Tamm form: τ_A is a characteristic evolution time for a chosen observable, not the standard deviation of an assumed universal time operator [6]. Other clock or quantum speed-limit statements require their own hypotheses.

7.3 Ordinary observable uncertainty

For self-adjoint A and B on a common state domain,

$$\Delta_\psi A \Delta_\psi B \geq \frac{1}{2} |\langle \psi, [A, B] \psi \rangle|.$$

This is a consequence of the Hilbert-space inner product and the supplied operators. It does not derive their commutator from an MTT projector. A complete MTT source theorem must identify the lower operators and prove that their commutator or Weyl relations descend from the same upper structure.

8 Three probability questions

The word “Born rule” can refer to three different achievements.

1. A *characterization theorem* says that a probability assignment satisfying specified additivity or noncontextuality assumptions has trace form.
2. An *operational source theorem* constructs a physical instrument and its output law from selected dynamics and a supplied quantum state.
3. A *pre-quantum or ontic theorem* explains why probability has that semantics before quantum states are accepted, or why one sample history is uniquely actual.

These are not equivalent.

8.1 What Gleason and Busch establish

Gleason’s theorem begins with a countably additive measure on the projection lattice of a Hilbert space of dimension at least three and concludes, under its hypotheses, that the measure has density-operator trace form [7]. Busch’s effect-algebra formulation similarly characterizes generalized probability assignments on effects [8]. These theorems are fundamental because they show how little freedom remains once Hilbert/effect structure and the relevant additivity assumptions are accepted.

They do not construct a detector, derive countable additivity from MTT closure, or explain why a physical preparation is represented by a positive normal functional. Consequently one cannot start with an arbitrary “re-coherence weight,” assume it is additive on every orthogonal partition, invoke Gleason, and then count the result as an independent MTT derivation of probability. The additivity assumption is already a major part of the probability contract.

8.2 Effects and instruments

A POVM E on an outcome space Ω assigns positive operators $E(B)$ with $E(\Omega) = I$. Given a density operator ρ , standard quantum probability is

$$\Pr_{\rho}(B) = \text{Tr}(\rho E(B)).$$

An instrument carries more information: each event B has a completely positive trace-nonincreasing map $\mathcal{I}_B$, with

$$\Pr_{\rho}(B) = \text{Tr} \mathcal{I}_B(\rho), \quad \rho_B = \frac{\mathcal{I}_B(\rho)}{\text{Tr} \mathcal{I}_B(\rho)}$$

when the denominator is nonzero. Measurement is therefore an ordinary physical interaction followed by a durable or readable record. It does not require consciousness or a fundamental observer boundary.

Naimark and Stinespring dilation theorems show that POVMs and completely positive maps admit larger-space representations [9, 10]. They do not say that the required ancilla, isometry, pointer algebra, and coupling are selected by MTT. Those remain source rows.

9 The selected $q = 79$ binary recorder

The present research corpus contains one result stronger than abstract Gleason characterization. It constructs a selected recorder on a declared binary apparatus domain and computes its full stopped output measure. The canonical theorem and its machine-readable certificate are owned by the MTT QM source-proof repository at commit `1615da7e1b2c917556fe04a44d073b905644071e` [11]. We summarize the result here without transferring the theorem's ownership.

9.1 Input data

The canonical-model source fixes more than a measurement effect [12]. Its finite carrier is the complexification of two real three-sheet permutation modules, with the selected shared-line complex structure. On the resulting complex six-dimensional fiber,

$$P = \text{diag}(J_3/3, J_3/3), \quad Q = I - P, \quad \text{rank } P = 2, \quad \text{rank } Q = 4,$$

where J_3 is the all-ones matrix. After the declared geometric and causal background hypotheses, this fiber supplies the Cauchy Hilbert space below, normal trace-class states, and the declared decomposable observable algebra. The Cauchy realization is not restricted to a finite-dimensional total Hilbert space. The finite symbol, rather than an arbitrarily inserted target potential, fixes the bounded internal reduced dynamics.

With one physical clock calibration $u = t/t_0$ and $\gamma = \log 448$, its channel is

$$\Phi_u(\rho) = P\rho P + Q\rho Q + 448^{-u}(P\rho Q + Q\rho P), \quad \mathcal{L}(\rho) = \gamma(P\rho P + Q\rho Q - \rho).$$

The semigroup, its minimal Fock dilation, and comparison of instruments are established at this canonical operational scope. The clock scale t_0 is metrological input, not a newly predicted number of seconds. Nor does fixing an unread channel fix an apparatus: distinct informative and state-independent phase-flip instruments can have the same channel. The selected binary output algebra and coupling are essential additional rows of this particular model.

Let

$$\mathcal{H}_{\Sigma} = L^2(\Sigma, d\mu_h; F_{q79}), \quad P = P_{\Sigma}, \quad Q = I - P,$$

with P and Q orthogonal. The selected one-anchor intrinsic clock has

$$\gamma = \log 448,$$

and the two recorder couplings are

$$L_p = \sqrt{\gamma} P, \quad L_q = \sqrt{\gamma} Q.$$

The environment is the two-channel symmetric Fock recorder with vacuum input. The selected preparation ensemble λ enters through its second moment

$$\rho_\lambda = \int |z\rangle\langle z| d\lambda(z).$$

The output number processes generate a commuting nondemolition algebra, so restriction of the selected normal joint state to that algebra has a classical spectral measure. This is standard quantum-stochastic machinery applied to MTT-selected projector, clock, preparation, and context data [13, 14, 15].

9.2 Exact output law

The no-count *amplitude* is $K_0(u) = e^{-\gamma u/2}I$, so its effect is $K_0(u)^*K_0(u) = e^{-\gamma u}I$. A line in the evolving status source calls the latter a propagator; it must be read as the effect, not as a Kraus amplitude [16]. The underlying output-measure proof has the half-exponent amplitude. This correction leaves the stopped probabilities unchanged.

For a horizon u , let r_u mean no count by u , and let (s, a) , $a \in \{p, q\}$, denote the first count at time s in channel a . The imported selected-source result gives the effects

$$F_u(r_u) = e^{-\gamma u}I, \quad F_u(ds, a) = \gamma e^{-\gamma s}P_a ds,$$

where $P_p = P$ and $P_q = Q$. Therefore

$$\mu_{\rho, u}(r_u) = e^{-\gamma u}, \quad \mu_{\rho, u}(ds, a) = \gamma e^{-\gamma s} \text{Tr}(\rho P_a) ds.$$

Conditional on a count by time u ,

$$\Pr_\rho(a \mid \text{count by } u) = \text{Tr}(\rho P_a).$$

No fitted probability vector and no independently appended classical Poisson process enter this selected output law.

The corresponding nonselective channel is

$$\Phi_u(\rho) = P\rho P + Q\rho Q + e^{-\gamma u}(P\rho Q + Q\rho P).$$

For two upper ensembles with the same second moment,

$$\rho_{\lambda_1} = \rho_{\lambda_2},$$

the complete stopped output measures agree. This is the exact second-moment capture descent proved on the canonical binary context.

At $u = 1$, the no-count probability is $1/448$. For the first selected carrier basis preparation, the conditional label weights are

$$(\Pr(p), \Pr(q)) = \left(\frac{1}{3}, \frac{2}{3}\right).$$

These numbers are outputs of the selected projector and clock in that context, not empirical fits.

9.3 What this closes and what it does not

The result closes the following statement:

Given standard normal-state operational semantics, the selected $q = 79$ binary P/Q recorder supplies its own stopped output measure and exact second-moment Born descent.

It does not derive the operational meaning of a normal state from a deterministic pre-quantum theory. A probability measure on the commuting output algebra is also not a distinguished character of that algebra and does not select one sample path as the uniquely actual history.

The universal source blocker therefore remains open for:

- arbitrary allowed apparatus contexts and outcome multiplicities;
- finite-bandwidth, detector-memory, and non-Markov corrections;
- a demanded pre-quantum derivation of probability semantics; and
- objective selection of one ontic history.

This boundary is not a defect in the exact binary theorem. It is the difference between a selected finite-domain result and a universal measurement theory.

9.4 Consumer boundary: continuum transport and ontology

The conditional continuum compiler is explained in the Born/record companion [17, 18]. A supplied positive self-adjoint continuum Hessian, invariant rank-three sector with kernel/support ranks one and two, and a selected isometry intertwining its projectors with the canonical pair transport the jump, drift, Fock cocycle, and stopped instrument when the same clock and minimal coupling are used. Approximate intertwining yields finite-horizon bounds, with a positive probability floor needed for normalized rare-event states. The physical endpoint, finite map, tail control, and clock still have to be selected. This conditional transport does not reopen the canonical Cauchy model.

The ontology result, contextualized in the Locality companion, concerns a different missing datum [19, 20]. The same operational record data admit single-record and coactual completions: the 1, 149, 298 partition of 448 atoms realizes the canonical checkpoint, and attaching the conditional states preserves its instrument. No actuality predicate is contained in the channel or dilation. This proves non-entailment, not a selected global hidden-variable law, a refutation of many-worlds, or universal apparatus control. Measurement remains ordinary record-forming physics under either operational description.

10 Composition, entanglement, and locality

Once the record supplies

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B,$$

entangled states are simply states that are not product states or mixtures of product states, according to the chosen purity level. Their existence is a consequence of the tensor-product state space. Projection may preserve, remove, or induce effective correlations depending on the embedding, but it does not by itself derive the tensor product or its physical subsystem meaning.

Local no-signalling likewise belongs to the operational algebra. If Λ_A is a trace-preserving completely positive map acting only on A , then for a bipartite density operator

$$\mathrm{Tr}_A[(\Lambda_A \otimes \mathrm{Id}_B)(\rho_{AB})] = \mathrm{Tr}_A \rho_{AB}.$$

Thus the statistics of B are unchanged by an unconditioned local operation on A . Conditioning on a communicated outcome can change the conditional state, but not produce superluminal signalling.

An MTT account may seek an upper-local explanation of Bell correlations. To establish it, the source map must still show which upper variables, settings, and record algebras descend to the lower tensor factors while preserving the relevant statistical-independence assumptions. That separate Bell-locality question is not solved by merely calling the upper space local.

11 Open systems and alternative representations

11.1 Weak-coupling master equations

Completely positive Markov semigroups have generators of Gorini–Kossakowski–Sudarshan–Lindblad form under the standard bounded or finite-dimensional hypotheses [21, 22]. Davies’ weak-coupling limit gives a rigorous route from a system–reservoir model to such a semigroup under spectral, correlation, scaling, and limiting assumptions [23]. These theorems do not imply that every MTT disturbance has a Markov limit.

The $q = 79$ recorder in Section 9 is a selected quantum-stochastic instrument on one declared domain. A general MTT open-system theorem would need to emit the reservoir state, interaction, correlation decay, renormalization, bandwidth regime, and approximation error. Memory effects are not failures of quantum mechanics; they indicate that a Markovian generator is not the correct reduced object.

11.2 Path integrals and semiclassical kernels

When a self-adjoint Hamiltonian and suitable action are already available, Trotter product formulas, Feynman–Kac formulas in imaginary time, or oscillatory integral constructions can provide kernel representations. Stationary phase can then recover classical trajectories and Van Vleck-type amplitudes in an appropriate semiclassical regime. These are alternative representations or asymptotic consequences of supplied dynamics. They do not select the Hamiltonian or establish a measure for an arbitrary real-time path integral [1, 24].

This distinction matters for MTT. A closure path or modal history can be a valuable upper description, but calling it a path integral does not make the quantum measure, phase, boundary condition, or continuum limit automatic.

12 Two concrete tests

12.1 A two-level recorder

Choose one normalized vector in $\mathrm{Ran} P$ and one in $\mathrm{Ran} Q$, and restrict the selected binary recorder to their span. In the resulting two-dimensional model, let

$$\mathcal{H} = \mathbb{C}^2, \quad P = |0\rangle\langle 0|, \quad Q = |1\rangle\langle 1|,$$

and prepare

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The canonical recorder formulas give

$$\mu_{\psi,u}(ds, p) = \gamma e^{-\gamma s} |\alpha|^2 ds, \quad \mu_{\psi,u}(ds, q) = \gamma e^{-\gamma s} |\beta|^2 ds.$$

Conditional on a count, the labels have weights $|\alpha|^2$ and $|\beta|^2$. This example is useful for two reasons. First, it shows exactly how the selected output instrument turns a coherent superposition into record statistics and dephasing. Second, the Hilbert space is two-dimensional, outside the direct scope of the original projection-lattice Gleason theorem. The result here comes from the instrument calculation, not from pretending that Gleason alone supplied the detector law.

12.2 A harmonic oscillator embedding

Let

$$H_{\text{osc}} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$$

on its standard self-adjoint domain. By Proposition 6.1, it can be placed as a reducing block of a larger upper Hamiltonian and recovered exactly on the coherent summand. The eigenvalues and propagator then agree with ordinary quantum mechanics.

This is an excellent decoder test: a wrong normalization, domain, or intertwiner is exposed immediately. It is not a prediction of m or ω . Those values were part of the inserted target block. To turn the example into a selected MTT result, an upstream source theorem must emit the oscillator scale and coupling without using the target spectrum as construction data.

13 From a geometric source to a retained operator

The following imports explain how the complete-record requirements can be met by a geometric source. They are not new local proofs of the source theorems. In particular, an exact construction acting on a *supplied* connection cannot select that connection by itself.

13.1 Projector realization and its domains

For a supplied compact unitary bundle with connection, the projective-HYM naturality result realizes the bundle as the range of a smooth finite-matrix projector $p(x)$, with an isometric module map U [25]. The intertwined differential and Hessian obey $D_p U = U D$ and $H_p = U H U^*$ on the range, with their metrics and operator domains transported. This is finite matrix size over an infinite-dimensional space of sections, not a finite Fourier truncation. Ambient vectors outside the range are not additional physical harmonic modes. Likewise, the external lane flag lives on a separate factor; it is not a flag of parallel subbundles inside an irreducible stable HYM bundle, whose parallel endomorphism commutant is scalar. The flat shared line and the curved bundle have different roles.

The explicit Cech compiler starts with Hermitian metrics and transitions, forms unitary transitions $u_{ij} = h_i^{1/2} g_{ij} h_j^{-1/2}$, and uses a partition satisfying $\sum_i \chi_i^2 = 1$. Its projector blocks are $p_{ij} = \chi_i \chi_j u_{ij}$ [26]. The bare Grassmann connection need not be the input connection. In a local realization,

$$A_0 = U^* dU, \quad \Gamma = U(A - A_0)U^*, \quad D_p = p d + \Gamma.$$

The correction is what makes the differential natural. Cover and partition changes give partial unitary equivalences on the range, not new physical source choices. Central phases cancel on an

adjoint bundle, but not on its twisted fundamental. Thus the ordinary deformation complex may have ranks $3 + 8 + 8 + 80 + 3 = 102$; it is not the finite 27-dimensional algebra carrier.

Cutoffs must respect this range identification. A simple winding isometry $U(x) = (3/5, (4/5)e^{ix})$ shifts the second Fourier component. Cutting both components at the same raw Fourier index loses squared norm $16/25$ on a boundary mode, although the uncut map is an exact isometry. Transporting the intrinsic finite subspace before applying the cutoff avoids confusing this presentation artifact with physical loss.

There is a separate dynamical test. A domain-preserving orthogonal projector R gives an exact compression only when $(I - R)HR = 0$. Otherwise the retained resolvent contains the Schur–Feshbach term of Section 5. In the source’s rational example, a retained mode of energy 1 couples by $1/2$ to an eliminated mode of energy 3. At spectral parameter zero its effective coefficient is $1 - (1/2)^2/3 = 11/12$, not 1. This illustrates why a closed pulled-back cost and a reducing isometry cannot be used interchangeably.

13.2 Strain coordinates and the cost of eliminating modes

For a Hermitian three-by-three matrix, write the diagonal coordinates as d_i , the oriented off-diagonal entries as z_{12}, z_{20}, z_{01} , and $r_e = |z_e|^2$. The relative phase torus leaves invariant not only d, r but also the triangle product

$$t = z_{12}z_{20}z_{01}, \quad (\operatorname{Re} t)^2 + (\operatorname{Im} t)^2 = r_{12}r_{20}r_{01}.$$

Thus the generic invariant quotient is seven-dimensional; the six strain coordinates $\Phi_{\text{strain}} = (d, r)$ deliberately forget a genuine triangle phase [27]. They are a nonlinear reduction, not an equivariant linear identification of the root plane with the two permutation modules. The respective S_3 characters $(6, 0, 0)$ and $(6, 2, 0)$ already prohibit that linear identification.

On the regular all-edge-nonzero stratum, the derivative $J = d\Phi_{\text{strain}}$ has rank six. Let H be the positive physical Hessian with a spectral gap on the declared harmonic complement and $G_{\perp} = H^{-1}$ its reduced Green operator. When the indicated covariance is invertible, the minimum upstairs cost compatible with a retained variation y is

$$\inf_{Jx=y} \frac{1}{2} \langle x, Hx \rangle = \frac{1}{2} \langle y, H_{\text{strain}} y \rangle, \quad H_{\text{strain}} = (JG_{\perp}J^*)^{-1}.$$

This is a shorted covariance, not the Hessian of an arbitrary linear slice. For a further TT sector the appropriate retained derivative and image must be used before taking the inverse. Inverting first and projecting later is not a general identity.

13.3 What changes when an edge vanishes

The support-stratified refinement supplies the boundary missing from a single regular-chart formula [28]. The eight edge-support masks have derivative ranks $3 + |S|$, with multiplicities 1, 3, 3, 1. The target is $\mathbb{R}^3 \times \mathbb{R}_+^3$. At an absent edge, the first derivative of $|z_e|^2$ vanishes and its second jet is $|\delta z_e|^2$. The tangential Green shorting remains valid on its proper image, but an outward normal strain η has the constrained value function

$$\inf \left\{ \frac{1}{2} \langle v, Hv \rangle : J_S v = 0, \ |v_e|^2 = \eta_e \text{ for absent edges} \right\}.$$

It is homogeneous of degree one in η , not an ordinary quadratic Hessian on six unrestricted boundary coordinates. For example, $r = t^2$ and cost $t^2/2$ give cost $r/2$, not $r^2/2$. This corrects the boundary consumer, not the regular-stratum theorem. A full rank-six derivative, or the proposed local full

six-coordinate complex symmetry, already forces the top stratum. Regularity is therefore not an additional independent obligation once that stronger input holds.

13.4 Monodromy, the shared root, and operator symmetry

A connected three-sheet cover with full S_3 monodromy is not an S_3 -Galois cover: its deck group is the centralizer of that monodromy in S_3 , hence trivial. The appropriate strain object is the associated bundle with fiber $\mathbb{R}^3 \oplus \mathbb{R}^3$, rather than six globally labeled functions [29]. Reynolds averaging gives parallel rank-two P and rank-four Q . The nonsplit binary lift has a central sign acting trivially on this quadratic strain representation, so its twelve-term average equals the six-term S_3 average. This does not settle a spinor's Spin or Spin^c structure.

If a closed operator, its domain, and the harmonic complement are monodromy-natural, the reduced Green operator is natural by uniqueness of the inverse. Symbol equivariance alone does not prove those hypotheses for a physical connection and Hessian.

The later shared-root refinement closes an important *symbol* question [30]. The two $\mathbb{Z}_{64}$ characters with exponents 1 and 33 restrict to the same i^m on the subgroup with elements 0, 16, 32, 48. The determinant-twisted sheet/edge identification then gives a parallel order-four map

$$J_{DE}(d, e) = (-e, d), \quad J_{DE}^2 = -I,$$

commuting with the S_3 action. On the two real standard TT copies an S_3 -invariant symmetric covariance has the form $\begin{pmatrix} a & c \\ c & b \end{pmatrix} \otimes I_2$. Commutation with J_{DE} forces $c = 0$ and $a = b$, leaving one positive scale. No independent TT shape fits are needed *after* this physical invariance is established.

The remaining issue is the actual nonzero-Chern HYM source and its Hessian, or a direct check of the physical TT block. The order-four map is not a global symmetry of the nonlinear positive strain orthant, nor an automorphism preserving the marked Fu–Yau branch: it exchanges the shared circle marking. A direct unital adjoint action on Hermitian matrices cannot implement a map sending their trace direction into an edge direction. These exclusions clarify how the source may realize the symbol; they do not revoke the canonical complex structure or the local order-four result.

13.5 Which source packages are still needed

The Fourier–Mukai/HYM dependency compression distinguishes compiler outputs from independent inputs [31]. In that literal route two of eleven rows are instantiated, while the nine uninstantiated rows factor through three sequential packages: a visible same-member spectral/flat-Deligne source, a common physical HYM endpoint, and physical order-four invariance or the direct TT block. WIT concentration, a locally free rank-three transform, determinant normalization, and the stated Chern-character output are consequences once their source hypotheses hold, not nine independent searches. A full Fourier–Mukai autoequivalence is an optional stronger route, not a prerequisite for every direct operator exit.

This compression preserves the hidden rank-nine projective construction and existential HYM theorem. It does not select a common visible–hidden endpoint or the visible norm-compatible twisted line. The physical support $9H + 3D_0$ is not the auxiliary $3H + 3D_0$ support. Vanishing integral Dixmier–Douady class is not holomorphic or flat-gerbe triviality; integral ambient quotients must retain their dual-lattice convention rather than replace $V^\vee$ by V . An ordinary Neron model is not automatic. Finite/formal HYM and closure-repair results therefore stay closed at their scopes without becoming a physical source selector.

The family-source result explains what a selected smooth family would buy [32]. Given a physical HYM/upper-action family with common Sobolev domains and a uniform spectral gap, one obtains

closed operators, Riesz projectors by contour integration, and Kato transport generated by $[\dot{P}, P]$. Determinant-line connection and parallel transport are then defined on that family. A gap crossing requires additional work; these constructions alone do not furnish a transverse fermion-measure current or disconnected phase gluing. Conversely, finitely many projected coefficients cannot select an infinite-dimensional smooth connection: the projection has a kernel. The packet’s rank-four projection on eight coordinates makes this elementary failure explicit with different lifts and different operators. It is a source-nonuniqueness witness, not a new physical HYM solution or a reopening of existential HYM.

14 The local field interface and its limits

14.1 A selected free net is already available

On the declared framed globally hyperbolic, spin-preserving category, the selected massless Dirac operator acts on spinor rank four tensored with the six-dimensional finite fiber, giving rank 24 [33]. Its causal propagator supplies the CAR algebra, graded locality, and the time-slice map; the even observables commute at spacelike separation. The commuting P/Q sectors have ranks 8 and 16, or 4 and 8 in each chirality. These are the accepted free-field results, conditional on the declared geometric/causal background, not obligations to restart. Omitting an added mass or gauge zero-order term defines this free tier; it does not predict all physical masses to vanish. The Hadamard state space is nonempty without selecting a preferred vacuum or identifying inequivalent spin structures.

One wording in the packet needs correction: quasifree states do not in general form a convex set. For two CAR modes, the equal mixture of vacuum and double occupation has $\langle n_1 \rangle = \langle n_2 \rangle = 1/2$ but $\langle n_1 n_2 \rangle = 1/2$, whereas Wick factorization with its vanishing cross-correlations gives $1/4$. Thus one may use the nonempty quasifree Hadamard class, or a convex ordinary Hadamard state space containing its convex hull; one must not call every member of that hull quasifree. Locality, positivity, and the free-net construction are unaffected.

14.2 Local quantum master equations are formal interacting results

The local QME result uses the accepted finite Standard Model content and declared profile-dependent coefficients on relatively compact on-shell globally hyperbolic charts in four dimensions [34]. Its interactions have compact support. Time-ordered local multilinear maps are controlled inside the equicausal functional setting; this is a domain condition, not permission to multiply arbitrary singular functionals. The five perturbative anomaly coefficients $SU(3)^3$, $SU(3)^2 U(1)$, $SU(2)^2 U(1)$, $U(1)^3$, and $\text{grav}^2 U(1)$ vanish, and the faithful $\mathbb{Z}_6$ spin-bordism obstruction is zero in the stated model. Causal renormalization, local BRST cohomology, and the Adler–Bardeen input remove exact remainders order by order. The result is an all-orders *formal* QME, nilpotent quantum BV differential, degree-zero physical algebra, and the stated gauge-fixing independence. No finite/profile Yukawa or electroweak result is being rederived here.

The evolving status source contains later mathematical refinements and cannot be discarded on the strength of its early open-status header [16]. On bounded charts with the specified $H^1 = 0$ condition, quartet reduction supplies formal positive physical states: the illustrative Gram matrix $\text{diag}(1, 1, 0)$ becomes positive on the physical quotient. State cones restrict contravariantly from a common parent. Equal restrictions on an overlap are necessary but not sufficient for arbitrary states to glue: two maximally entangled states on AB and BC have the same maximally mixed B

marginal but no common *ABC* extension. This is ordinary quantum marginal compatibility, not a failure of the formal local algebra or a signal between regions.

14.3 What the later regulator refinements actually control

The same status source develops compact Hilbert-BV Hodge projectors and finite-shell free cycles, with Kato transport under common-boundary and gap hypotheses. Collar and ambient-isotopy changes concern presentation of the declared local model. The auxiliary sign-flipped metric $g_E = g_L + 2n^b \otimes n^b$ organizes an elliptic construction; it is not a physical Wick-rotation theorem or a global reflection symmetry. Even matching a slice is insufficient for the latter.

For counterterms the first-order Yang–Mills complex has ranks $1 \rightarrow 7 \rightarrow 7 \rightarrow 1$; its heat-kernel construction must not be replaced by a naive quartic detour complex. A direct Lorentzian alternative smooths Cauchy data with $R_\varepsilon = \exp[-\varepsilon(\log 2)A_\Sigma]$. At finite cutoff this smoothing is nonlocal and satisfies modified Ward identities. After local subtraction, the normalized causal-renormalization bridge recovers the formal QME. These are genuine local regulator and comparison advances, not a positive interacting continuum limit at fixed coupling. The older status numbers attached to intermediate routes are not current global completion counts.

14.4 Free positivity and fixed-coupling nonpromotion

The free C^* construction combines the physical gauge Weyl algebra, four-real-component Higgs Weyl algebra, and rank-48 chiral CAR factor with the minimal tensor product, then takes the declared compact-gauge and even observables [35]. Its state/GNS and local von Neumann constructions are established at zero coupling. Quasifree local quasiequivalence applies to the stated Higgs/fermion sectors; for the gauge factor the comparison stays inside each constraint-preserving, implementable Hilbert–Schmidt orbit. It is not an arbitrary global gauge orbit theorem.

There are also literal finite interacting regulator algebras at declared nonzero coupling and a norm-Duhamel first-tangent comparison with the formal theory. The comparison transports observables through the renormalization map: a normalized first derivative does not erase its higher derivatives on composite observables. Orbitwise normalized finite chiral determinants remove a common absolute phase, but relative phases across disconnected pieces, transverse currents, crossings, and uniform limiting control are separate data. A determinant line is not by itself a full fermion measure.

The packet gives an exact reason that agreement to every perturbative order does not select a smooth fixed-coupling completion. Set

$$f(\lambda) = \frac{1}{2}e^{1-1/\lambda^2} \quad (\lambda \neq 0), \quad f(0) = 0.$$

All its derivatives at zero vanish, but $f(1) = 1/2$. Hamiltonians with diagonal entries $(0, 1)$ and $(f(\lambda), 1)$, with correspondingly different positive normalized states, can therefore share every formal jet and differ at finite coupling. This excludes inference from smooth formal jets alone. It does not exclude a future analytic, summability, or genuinely nonperturbative source theorem. Free positivity, formal interacting QME, and finite-regulator positivity remain distinct established statements.

15 Relation to the current MTT program

The canonical MTT foundation now uses a ten-dimensional bundle over a four-dimensional Lorentzian base with compact six-dimensional fiber, not a literal product of three independent internal manifolds [4]. The shared finite line and root-plane complex structure already supply the canonical complex

operational sector. Matching dimensions or finite carriers alone does not supply the additional physical HYM-to-continuum operator-intertwining theorem.

The current authority distinguishes established canonical QM, free-field, and local formal constructions from their genuine source and interacting continuum extensions. The present paper uses that distinction:

1. *General coherent-sector QM*: conditional reconstruction.
2. *Canonical finite-symbol Cauchy QM*: established operational model, reduced semigroup, minimal dilation, and instrument comparison.
3. *Selected binary output law*: exact on the canonical $q = 79$ recorder domain.
4. *Universal Born source*: open.
5. *Selected free CAR net and local formal QME*: established at their declared free and formal domains.
6. *Nonperturbative QFT and continuum physics*: separate, open obligations not promoted here.

The possible upstream “closure repair” or nonlinear stabilization program would fit one level before this paper. If a selected nonlinear repair flow has a fixed state and its linearization produces the self-adjoint or dissipative operators in Definition 2.1, then the operator row would become a theorem rather than an input. That is a promising research direction, but no such universal source theorem is assumed in the present reconstruction.

16 Claim disposition

Complete derivation

Withdrawn. The general result is a conditional coherent-sector reconstruction.

Hilbert structure

Must be emitted as a complex Hilbert space or as a real Hilbert space with a compatible selected complex structure. It is not derived from projection alone.

Self-adjoint Hamiltonian

Retained when a selected densely defined closed semibounded form or self-adjoint operator and its domain are proved. Bounded pullback preserves closedness on the full preimage domain, but alone supplies neither density nor a reducing operator part.

Unitary dynamics

Retained as a standard consequence of self-adjointness and Stone’s theorem. Time-dependent propagators remain conditional on their regularity hypotheses.

Arbitrary Schrodinger models

Retained as exact representability when the target form is inserted. This is not source selection or parameter prediction.

Noncommutativity

Part of the supplied operator algebra or a separate source theorem. It is not created by an orthogonal projector.

Born rule	Gleason–Busch gives characterization under probability assumptions. Selected exact Born descent is established only for the canonical binary $q = 79$ output context. The universal source theorem remains open.
POVMs and dilations	Standard mathematical representations. MTT must separately source the ancilla, interaction, pointer algebra, and context.
Time uncertainty	Corrected to an operational clock or Mandelstam–Tamm statement. No universal self-adjoint time operator is claimed.
Entanglement	Available once the tensor-product composition rule is supplied. Projection alone does not derive physical subsystem composition.
Lindblad dynamics	Conditional on an explicit open-system or weak-coupling limit. Not every modal disturbance is Markovian.
Path-integral language	An alternative or asymptotic representation after the Hamiltonian and action are supplied, not an independent source theorem.

17 Completion contract

To upgrade the general reconstruction to a selected derivation, one upper source must emit and certify the following rows without empirical replay of the desired answer:

1. one complex coherent Hilbert space and its state cone;
2. one noncommutative observable algebra with domains and symmetries;
3. one self-adjoint Hamiltonian or closed semibounded form, including $\hbar$, masses, couplings, and boundary conditions;
4. a proof that the coherent range reduces the dynamics, or a controlled Schur–Feshbach/residual theorem when it does not;
5. a clock convention and physical calibration distinct from a compact phase circle unless an explicit relation is proved;
6. a tensor-product or algebraic composition rule with locality;
7. selected effects, instruments, record algebras, and apparatus couplings;
8. a same-source output-measure theorem for the entire allowed context family, including finite-bandwidth and memory error control;
9. any requested pre-quantum probability semantics or objective-history rule; and
10. a commuting evaluation diagram with held-out predictions and an uncertainty budget.

The canonical $q = 79$ model instantiates its own Hilbert, state, observable, clock, semigroup, dilation, and instrument rows, including item 8 on one binary context. It does not close this more general all-apparatus and physical-source contract by implication. Conversely, future progress on the upper action, operator naturality, or closure-repair flow could source several earlier rows at once. The contract makes such progress visible without changing the meaning of already closed results.

18 Discussion

18.1 What has actually been achieved

MTT can host a mathematically exact coherent-sector reconstruction of nonrelativistic quantum mechanics when the full lower record is supplied and the source/evaluation diagrams commute. The form and compression analysis specifies when the Hamiltonian is genuinely self-adjoint and when coherent evolution is exact. The target-embedding result establishes broad expressiveness while explicitly separating it from prediction.

More importantly, the current selected $q = 79$ program goes beyond a bare encoding on one measurement domain. It fixes a projector pair, intrinsic rate, recorder unitary, commuting output algebra, and stopped instrument. The resulting first-count law depends only on the preparation's second moment and has exact trace-form label probabilities. No observed probabilities or free stochastic noise are added. This is a real selected-source achievement, but its domain matters.

18.2 Why the narrower statement is stronger science

The former version appeared stronger because it called many standard theorems MTT derivations. In fact, it made the decisive source question harder to see. The corrected statement is more useful: it identifies which structures are standard consequences, which are MTT-selected in a finite domain, and which remain open.

The distinction also prevents two opposite mistakes. It avoids declaring victory after replaying a target Hamiltonian, and it avoids discarding a genuine finite-domain theorem merely because universality is unfinished. Scientific progress can be exact and important without being global.

18.3 Best next theorem

The immediate mathematical target is not another abstract Gleason argument. It is a same-source extension of the selected recorder theorem. One should enlarge the apparatus class, derive the corresponding output algebra and instrument from the same upper source, and prove second-moment capture descent with explicit control of detector bandwidth and memory. In parallel, a nonlinear closure-repair flow whose linearization emits the Hamiltonian and recorder couplings would move the source boundary further upstream.

19 Conclusion

Projection is not quantization, and reconstruction is not selection. Nonrelativistic quantum mechanics requires a complex Hilbert space, state cone, operator algebra, self-adjoint dynamics, clock convention, composition rule, and operational instruments. This revision makes each of these rows explicit and proves the corresponding conditional MTT reconstruction theorem.

The operator-theoretic part is rigorous once its standard hypotheses are met. Closed reducing forms give self-adjoint coherent Hamiltonians; Stone's theorem gives unitary evolution; and arbitrary

target forms can be embedded exactly. That last fact measures expressive capacity, not predictive power. Time is treated through evolution or clock observables, and the valid Mandelstam–Tamm relation replaces an unsupported universal time operator. Gleason and Busch characterize probability assignments but do not source a detector law.

The canonical $q = 79$ model supplies a selected operational realization, including an exact stopped output measure and second-moment Born descent on one commuting Fock output domain, without fitted probabilities or extra classical noise. The universal apparatus theorem, non-Markov control, pre-quantum probability semantics, and objective actualization remain open. MTT therefore reaches a conditional reconstruction of general nonrelativistic quantum mechanics and an exact selected measurement theorem on one important domain. The projector, stratified-strain, and family compilers explain the remaining physical-source interface; the free net and local formal QME remain positive results without implying an interacting fixed-coupling continuum completion.

Frozen contextual sources for Version 5

The thirteen contextual imports cited above use the immutable results manifest at commit `f141a20ea23c5c3f f19cc216 1c0e226e 29ade8a7`. Each artifact was read in full, including later refinements in the evolving status source, and its frozen byte hash is recorded in the accompanying review. The local revision audit distinguishes source reading and bounded illustrative checks from independent verification of the source theorems. Current authority determines scope; early packet status headings do not override later proof explanations. The earlier ledger provenance below remains historical provenance for that separately cited row.

Computational Evidence and Reproducibility

The curated strict-upgrade ledger is used only as a corpus-state cross-check that stronger no-knob and universal-source obligations remain open. It does not prove this paper’s Hilbert, Hamiltonian, clock, instrument, or probability statements. The exact canonical $q79$ binary output-measure theorem is instead cited at its immutable source-proof repository commit and remains restricted to its declared P/Q Fock-output domain.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).

Historical 2/9 strict no-knob ledger snapshot, not a current global completion count.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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